Understanding the Relationship Between Rodent Violations and Extreme Temperatures

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Research Question:

What is the relationship between the number of rodent-related health code violations and the frequency of heat advisory warnings (extreme summer temperatures) in Boston from 2007-2025?

Background:

- Increase of extreme temperatures during summer due to climate change.
- Rats actively avoid extreme heat, seeking shelter in human spaces.
- This causes restaurants to be at risk, as they have an abundance of food, water, and shelter.

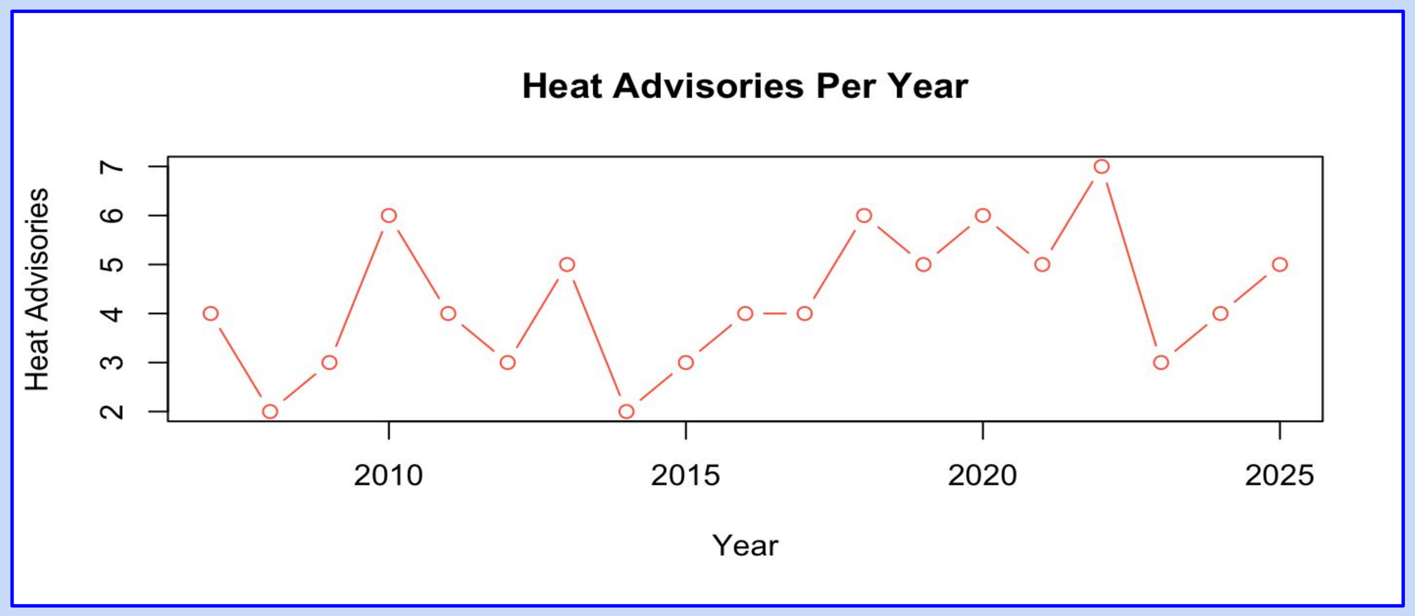

Figure 1: Scatterplot of heat advisories over time

Data & Methods:

Primary Data Source:

- Boston Food Inspections; **Attributes**: violation time, comments
- NOAA Climate Data for Boston ; **Attributes**: max temp

Data Manipulation:

- Created rodent_rate (total violations in a day / total inspections in a day)
  - Characterized by “rodent” in comments.
- Created a heat wave start variable (if it was the beginning of a heat advisory – 3 days of 85° or more).
- If the number of violations were outliers, that observation was removed.
- Merged these two datasets using the “date” column as a primary key.
- If a food inspection row did not contain the violation time, it was dropped.
- For the logistic regression, any_rodent was created to signify if a rodent violation was found on a day.
- For the beta regression, only dates with rodent rates that were greater than 0 were included.
- 80% train, 20% test split

Visualizations:

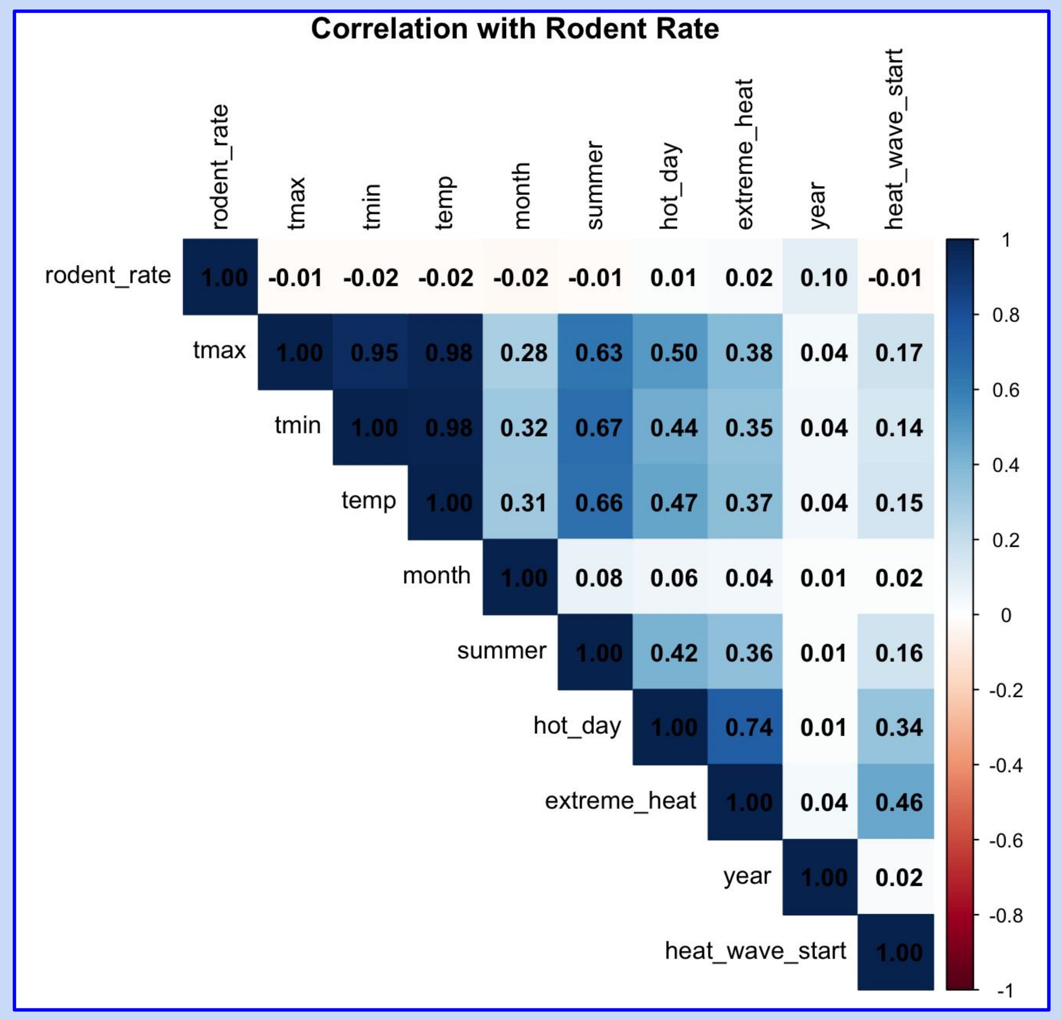

Figure 2: Correlation plot of all potential variables.

- Relationship between our response (rodent rate) and our explanatory variables are very weak.
- The strongest correlation is between rodent rate and year, or a measure of time.
- The heat related variables do not show strong correlation at all.

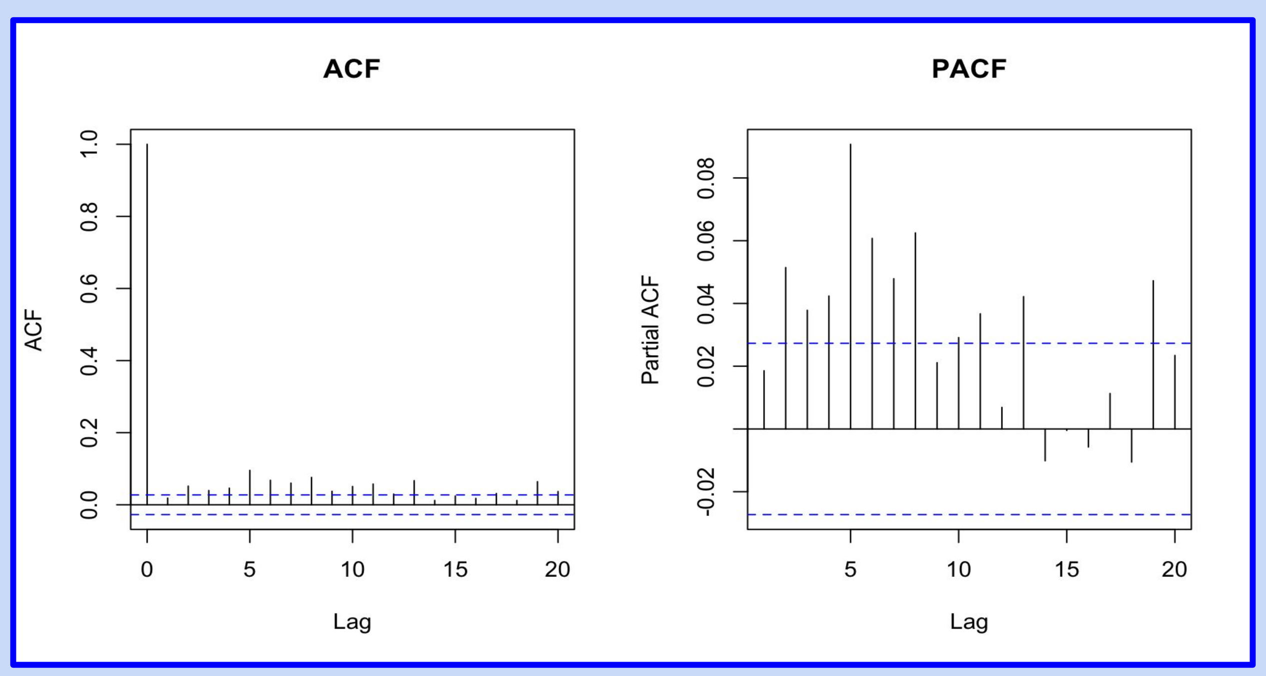

Figure 3: ACF and PACF plots

Modeling:

- **ACF** measures correlation between a time series and its lagged values at different time lags
- Shows **total correlation** (both direct and indirect effects)
- minimal autocorrelation across lags (weak dependence between lagged data and current)
- **PACF** measures correlation between a time series and its lag, **without** influence of intermediate lags
- Shows **direct correlation only** (filters out indirect effects)
- Compared two models: with lag 5 term vs. without lag term (baseline)
- AIC scores differed by less than 0.01 (negligible improvement)

Model 1: OLS Regression with no autocorrelation
Model 2:
Part 1: Logistic Regression – Any rodent violations on a specific day?
Part 2: Beta Regression – Given that violations actually occurred, what rodent rate can be predicted?

Results:

	Model 2, Pt. 1: Logistic Regression	Model 2, Pt. 2: Beta Regression
Date	0.0001 p-value: < 0.001 ***	0.0083 p-value: 0.378
Max Temperature	-0.0099 p-value: < 0.001 **	-0.0012 p-value: 0.201
Heat Wave Start?	.0119 p-value: 0.97038	-0.0559 p-value: 0.546
Combined RMSE	0.0848	

	Logistic Regression	Beta Regression
Regressions with and without heat variables (ANOVA results)	p-value: < 0.01 ***	p-value: 0.3154

Logistic Regression (Part 1): p = 0.0099***
- Highly significant → Heat variables are helpful for predicting whether you get a zero.

Beta Regression (Part 2): p = 0.3154
- Not significant →Heat variables don't matter for predicting the magnitude among non-zeros

** On average, predictions are off by ~0.085 in rodent rate

Conclusion: -

- Heat advisories **do not appear** to associate with the increase rodent violation rates in Boston from 2007 to 2025
- Notable: The **negative max temp coefficient in Part 1**; This suggests that inspections may be suppressed on hot days.
- Appears that for reaching the threshold to have a rodent violation, heat does play a role.
  - However, when determining the magnitude of the rate, heat does not appear to be correlated.

Limitations

- Exclusion of dates – outliers
  - Data quality issues?
  - Detection != violation occurrence time
- Potentially missing variables – only have temp related and date variables

Discussion & Future Work

- Contrasts the existing research done on rodent sightings and extreme temperature.
  - Heat advisories may not be he only thing to be focusing on (humidity, rain)
- Insinuates the heat correlates with humans behavior rather than rodents